Simplex quadrature nodal DG method for wave equations on sphere

April 25, 2026

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1. $2n-1$ and $2n$ Quadrature points in 2-simplex

1.1. Barycentric Coordinates

Consider the two-dimensional plane with coordinates $\boldsymbol{\xi} = (\xi, \eta)$. Let T be the triangle with vertices

$$\boldsymbol{\xi}_{v_1} = (0,1), \quad \boldsymbol{\xi}_{v_2} = (0,0), \quad \boldsymbol{\xi}_{v_3} = (1,0)$$

Any point within the triangle can be expressed as

$$(\xi, \eta) = \boldsymbol{\xi} = \sum_{i=0}^{3} b_i \boldsymbol{\xi}_{v_i}, \quad \sum_{i=1}^{3} b_i = 1, \quad b_i \geq 0 \tag{1}$$

where b_i for $i = 1, 2, 3$ are known as the barycentric coordinates.

Numerical integration is often encounter in computations. To conduct numerical integration over T, numerical quadrature points and weights are introduced in []. Below we quote the barycentric coordinates of the quadrature points and the associated weights for conducting integration of polynomials of degree at most degree $2n-1$ and $2n$, in Table 1 and Table 2.

n	Sym	b_1	b_2	w_i	w_i^b
1	S_6	0.2113248654051871	0.0000000000000000	0.16666666666666667	0.5000000000000000
2	S_6	0.1127016653792583	0.0000000000000000	0.04166666666666666	0.2777777777777777
2	S_3	0.5000000000000000	0.0000000000000000	0.09999999999999999	0.4444444444444444
2	S_1	0.3333333333333333	0.3333333333333333	0.45000000000000000	–
3	S_6	0.06943184420297367	0.0000000000000000	0.01509901487256561	0.1739274225687269
3	S_6	0.3300094782075718	0.0000000000000000	0.04045654068298990	0.3260725774312731
3	S_6	0.5841571139756568	0.1870738791912763	0.11111111111111111	–
4	S_6	0.04691007703066797	0.0000000000000000	0.006601315081001592	0.1184634425280944
4	S_6	0.2307653449471584	0.0000000000000000	0.02053045968042892	0.2393143352496833
4	S_3	0.5000000000000000	0.0000000000000000	0.01853708483394990	0.2844444444444446
4	S_3	0.1394337314154536	0.1394337314154536	0.10542932962084440	–
4	S_3	0.4384239524408185	0.4384239524408185	0.12473673228977350	–
4	S_1	0.3333333333333333	0.3333333333333333	0.09109991119771331	–

Table 1: Barycentric coordinates of quadrature integration points for integrating polynomials of degree at most $2n-1$ in a triangle. w_i are surface integration associated quadrature weights. w_i^b are line integration quadrature weights associated with edge points.

n	Sym	b_1	b_2	w_i^s
1	S_3	0.1666666666666666	0.1666666666666666	0.3333333333333333
2	S_3	0.09157621350977067	0.09157621350977067	0.1099517436553218
2	S_3	0.4459484909159648	0.4459484909159648	0.2233815896780115
3	S_3	0.219429982549783	0.219429982549783	0.1713331241529809
3	S_3	0.480137964112215	0.480137964112215	0.08073108959303095
3	S_6	0.1416190159239682	0.0193717243612408	0.04063455979366068
4	S_6	0.7284923929554044	0.2631128296346379	0.02723031417443505
4	S_3	0.4592925882927232	0.4592925882927232	0.09509163426728455
4	S_3	0.1705693077517602	0.1705693077517602	0.1032173705347182
4	S_3	0.05054722831703096	0.05054722831703096	0.03245849762319804
4	S_1	0.3333333333333333	0.3333333333333333	0.1443156076777874

Table 2: Barycentric coordinates of quadrature integration points for integrating polynomials of degree at most $2n$ in a triangle. w_i^s are surface integration associated quadrature weights.

Based on these points we evaluate surface integration over a triangle, with $|T|$ being the area of the triangle, by a discrete summation as follows,

$$\int_T f(\boldsymbol{\xi})\,d\boldsymbol{\xi} = |T|\sum_{i=1}^{N} f(\boldsymbol{\xi}_i) w_i^s, \tag{2}$$

provided that f is a polynomial of degree at most $2n-1$ and $2n$. For evaluating line integrals on an edge of a triangle, we adopt the following quadrature formula:

$$\int_{\partial T^\gamma} f(\boldsymbol{\xi})\,d\boldsymbol{\xi} = |\partial T^\gamma|\sum_{i=1}^{n+1} f(\boldsymbol{\xi}_i^\gamma) w_i^e \tag{3}$$

where $|\partial T^\gamma|$ denotes the length of the edge γ, $\boldsymbol{\xi}_i^\gamma$ are the quadrature points on the edge, and w_i^e are the associated quadrature weights.

2. Orthogonal Polynomials on Simplex

We denote the triangle with vertices $(-1,1)$, $(-1,-1)$, and $(1,-1)$ by

$$\mathsf{T}^2 = \{(r,s)\,|\,r,s \geq -1,\, r+s \leq 1\}.$$

Let $n \in \{1,2,3,4\}$. On T^2, let us introduce the orthogonal polynomials $\phi_{i,j}$ defined by

$$\phi_m(r,s) = \Phi_{ij}(r,s) = \sqrt{2}P_i(a)P_j^{(2i+1,j)}(b)(1-b)^j, \quad 0 \leq i,j \leq n, \quad i+j \leq n,$$
$$m = j + (n+1)i + 1 - \frac{i}{2}(i-1), \quad a = 2\cdot\frac{1+r}{1-s} - 1, \quad b = s$$

where P_i is the Legendre polynomial of degree i and $P_j^{(2i+1,j)}$ is the Jacobi polynomials of degree j. Applying these basis functions we span the function space as

$$\mathcal{V} = \text{span}\{\Phi_{ij}(r,s) \,|\, 0 \le i,j \le n, \quad i+j \le n\}$$

Thus, for $n = 1$, we have the basis functions

$$\begin{aligned}
\frac{\phi_1(r,s)}{\sqrt{2}} &= \frac{\Phi_{00}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_0^{(1,0)}(s)(1-s)^0 = 1 \\
\frac{\phi_2(r,s)}{\sqrt{2}} &= \frac{\Phi_{01}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_1^{(1,1)}(s)(1-s)^1 = 2 + 4\frac{s-1}{2}(1-s)^1 = 2 - 2(1-s)^2 \\
\frac{\phi_3(r,s)}{\sqrt{2}} &= \frac{\Phi_{10}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_0^{(3,0)}(s)(1-s)^0 = 1 + \left(\frac{2(1+r)}{1-s} - 1\right) = 2\frac{1+r}{1-s}
\end{aligned}$$

For $n = 2$, we have six functions,

$$\begin{aligned}
\frac{\phi_1(r,s)}{\sqrt{2}} &= \frac{\Phi_{00}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_0^{(1,0)}(s)(1-s)^0 = 1 \\
\frac{\phi_2(r,s)}{\sqrt{2}} &= \frac{\Phi_{01}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_1^{(1,1)}(s)(1-s)^1 = 2 - 2(1-s)^2 \\
\frac{\phi_3(r,s)}{\sqrt{2}} &= \frac{\Phi_{02}}{\sqrt{2}} = P_0\left(\frac{2(1+r)}{1-s}\right) P_2^{(1,0)}(s)(1-s)^2 = \left(3 + 6\left(\frac{2(1+r)}{1-s} - 1\right) + \frac{5}{2}\left(\frac{2(1+r)}{1-s} - 1\right)^2\right)(1-s)^2 \\
\frac{\phi_4(r,s)}{\sqrt{2}} &= \frac{\Phi_{10}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_0^{(3,0)}(s)(1-s)^0 = 2\frac{1+r}{1-s} \\
\frac{\phi_5(r,s)}{\sqrt{2}} &= \frac{\Phi_{11}}{\sqrt{2}} = P_1\left(\frac{2(1+r)}{1-s}\right) P_1^{(3,1)}(s)(1-s)^1 = 2\frac{1+r}{1-s}\left(4 + 3\left(\frac{2(1+r)}{1-s} - 1\right)\right)(1-s)^1 \\
\frac{\phi_6(r,s)}{\sqrt{2}} &= \frac{\Phi_{20}}{\sqrt{2}} = P_2\left(\frac{2(1+r)}{1-s}\right) P_0^{(5,0)}(s)(1-s)^0 = 1 + 3\left(\frac{2(1+r)}{1-s} - 1\right) + \frac{3}{2}\left(\frac{2(1+r)}{1-s} - 1\right)^2
\end{aligned}$$

3. Approximation, Integration, and Differentiation

3.1. Modal and nodal approximation

Based on these grid points, we may seek numerical approximations to a function u defined on T.

Let $\boldsymbol{\xi} = (\xi, \eta)$. Consider a function space $\mathcal{V}$ spanned by a set of linearly independent basis functions $\phi_k(\boldsymbol{\xi})$ for $i = 1, 2, \cdots, n$, i.e.,

$$\mathcal{V} = \text{span}\{\phi_i(\boldsymbol{\xi}) | i = 1, 2, ..., n\}.$$

Thus, a function $u(\boldsymbol{\xi}) \in \mathcal{V}$ can be approximated as follows,

$$u(\boldsymbol{\xi}) = \sum_{i=1}^{N} a_i \phi_i(\boldsymbol{\xi}), \tag{4}$$

where a_i are coefficients. Applying the quadrature points $\boldsymbol{\xi}_m = (\xi_m, \eta_m)$ for $m = 1, 2, \cdots, M$ and denoting $u(\boldsymbol{\xi}_m) = u_m$ for simplicity, we have

$$\boldsymbol{u} = \boldsymbol{V}\boldsymbol{a}, \tag{5}$$

where

$$\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_M \end{bmatrix}, \quad \boldsymbol{V} = \begin{bmatrix} \phi_1(\boldsymbol{\xi}_1) & \phi_2(\boldsymbol{\xi}_1) & \cdots & \phi_N(\boldsymbol{\xi}_1) \\ \phi_1(\boldsymbol{\xi}_2) & \phi_2(\boldsymbol{\xi}_2) & \cdots & \phi_N(\boldsymbol{\xi}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\boldsymbol{\xi}_M) & \phi_2(\boldsymbol{\xi}_M) & \cdots & \phi_N(\boldsymbol{\xi}_M) \end{bmatrix}, \quad \boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix}. \tag{6}$$

Let

$$\boldsymbol{W} = \mathrm{diag}(w_1, w_2, \cdots, w_M) \tag{7}$$

Define the mass matrix over the triangle T as

$$\boldsymbol{M} = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}. \tag{8}$$

Since $\boldsymbol{W}$ is diagonal, we have from Eq. (8)

$$\boldsymbol{M}^T = |T|(\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V})^T = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V} = \boldsymbol{M} \tag{9}$$

Furthermore, for $\boldsymbol{a}$ being a non-zero vector and $\boldsymbol{W}$ being positive definite, we have

$$\boldsymbol{a}^T\boldsymbol{M}\boldsymbol{a} = |T|(\boldsymbol{V}\boldsymbol{a})^T\boldsymbol{W}(\boldsymbol{V}\boldsymbol{a}) > 0. \tag{10}$$

indicating that $\boldsymbol{M}$ is not only symmetric but also positive definite. Thus, its inverse exists, denoted by $\boldsymbol{M}^{-1}$.

Then, multiplying $|T|\boldsymbol{V}^T\boldsymbol{W}$ to Eq. (5), we have the vector $\boldsymbol{a}$ as follows,

$$|T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{u} = |T|\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}\boldsymbol{a} = \boldsymbol{M}\boldsymbol{a} \implies \boldsymbol{a} = |T|\boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{u} \tag{11}$$

3.2. Numerical differentiation

To perform numerical differentiation on the vector $\boldsymbol{u}$, we need to construct the differentiation matrices for differentiating functions in the ξ and η directions. Differentiating Eq. (4) with respect to ξ, we have

$$\partial_\xi u(\boldsymbol{\xi}) = \sum_{i=1}^{N} a_i \partial_\xi \phi_i(\boldsymbol{\xi}) \tag{12}$$

Hence, the grid vector version of the above expression is

$$\boldsymbol{u}_\xi = \boldsymbol{V}_\xi\boldsymbol{a}, \quad \boldsymbol{u}_\xi = \begin{bmatrix} \partial_\xi u_1 \\ \partial_\xi u_2 \\ \vdots \\ \partial_\xi u_M \end{bmatrix}, \quad \boldsymbol{V}_\xi = \begin{bmatrix} \partial_\xi\phi_1(\boldsymbol{\xi}_1) & \partial_\xi\phi_2(\boldsymbol{\xi}_1) & \cdots & \partial_\xi\phi_N(\boldsymbol{\xi}_1) \\ \partial_\xi\phi_1(\boldsymbol{\xi}_2) & \partial_\xi\phi_2(\boldsymbol{\xi}_2) & \cdots & \partial_\xi\phi_N(\boldsymbol{\xi}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_\xi\phi_1(\boldsymbol{\xi}_M) & \partial_\xi\phi_2(\boldsymbol{\xi}_M) & \cdots & \partial_\xi\phi_N(\boldsymbol{\xi}_M) \end{bmatrix} \tag{13}$$

Applying the relationship Eq. (11) we have

$$\boldsymbol{u}_\xi = \boldsymbol{V}_\xi \boldsymbol{a} = |T|\boldsymbol{V}_\xi \boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{u} = \boldsymbol{D}_\xi \boldsymbol{u} \tag{14}$$

where $\boldsymbol{D}_\xi$ is the differentiation matrix in the ξ direction. Likewise, we construct the differentiation matrix in the η direction and perform the associated numerical differentiation as follows,

$$\boldsymbol{V}_\eta = \begin{bmatrix} \partial_\eta\phi_1(\boldsymbol{\xi}_1) & \partial_\eta\phi_2(\boldsymbol{\xi}_1) & \cdots & \partial_\eta\phi_N(\boldsymbol{\xi}_1) \\ \partial_\eta\phi_1(\boldsymbol{\xi}_2) & \partial_\eta\phi_2(\boldsymbol{\xi}_2) & \cdots & \partial_\eta\phi_N(\boldsymbol{\xi}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_\eta\phi_1(\boldsymbol{\xi}_M) & \partial_\eta\phi_2(\boldsymbol{\xi}_M) & \cdots & \partial_\eta\phi_N(\boldsymbol{\xi}_M) \end{bmatrix}, \quad \boldsymbol{D}_\eta = |T|\boldsymbol{V}_\eta \boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{W}, \quad \boldsymbol{u}_\eta = \boldsymbol{D}_\eta \boldsymbol{u}. \tag{15}$$

3.3. Numerical Integration

Here we briefly summarize the extension of the quadrature rule.

Let $q(\boldsymbol{\xi})$ be a scalar function and $\boldsymbol{V}(\boldsymbol{\xi}) = (u(\boldsymbol{\xi}), v(\boldsymbol{\xi}))$ be a vector-valued function. From the divergence theorem, we have

$$\oint_{\partial\mathcal{T}} \boldsymbol{n}\cdot(q\boldsymbol{F})\,d\boldsymbol{\xi} = \int_{\mathcal{T}} \nabla\cdot(q\boldsymbol{F})\,d\boldsymbol{\xi} = \int_{\mathcal{T}} q\nabla\cdot\boldsymbol{F}\,d\boldsymbol{\xi} + \int_{\mathcal{T}} \boldsymbol{F}\cdot\nabla q\,d\boldsymbol{\xi} \tag{16}$$

If both q and $\boldsymbol{F}$ are polynomials of degree at most n then we can apply the simplex quadrature rule to establish discrete version of the equation as follows,

$$\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{f}_b = |T|\boldsymbol{1}^T\boldsymbol{W}\left(\boldsymbol{D}_\xi(\boldsymbol{q}\odot\boldsymbol{u}) + \boldsymbol{D}_\eta(\boldsymbol{q}\odot\boldsymbol{v})\right) \tag{17}$$

$$= |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{D}_\xi\boldsymbol{u} + \boldsymbol{D}_\eta\boldsymbol{v}) + |T|(\boldsymbol{u}^T\boldsymbol{W}\boldsymbol{D}_\xi\boldsymbol{q} + \boldsymbol{v}^T\boldsymbol{W}\boldsymbol{D}_\eta\boldsymbol{q}) \tag{18}$$

where $\boldsymbol{q}_b$ and $\boldsymbol{f}_b$ are grid vector representations of the functions q and $\boldsymbol{n}\cdot F$ at the three edges, and $\boldsymbol{W}_b$ is the quadrature weight matrix with diagonal entries being w_i^b for $i = 1, 2, \ldots, n+1$,

$$\boldsymbol{W}_b = \begin{bmatrix} |\partial T^1| & 0 & 0 \\ 0 & |\partial T^2| & 0 \\ 0 & 0 & |\partial T^3| \end{bmatrix} \otimes \mathrm{diag}(w_1^b, w_2^b, \ldots, w_{n+1}^b)$$

3.4. Well-posed analysis for model wave equations

Let us consider applying discontinuous Galerkin method to solve the model wave equation:

$$\partial_t q(\boldsymbol{\xi}, t) + \nabla\cdot(\boldsymbol{V}q(\boldsymbol{\xi}, t)) = 0, \quad \boldsymbol{\xi}\in\mathcal{T} \tag{19}$$

$$q(\boldsymbol{\xi}, 0) = q_0(\boldsymbol{\xi}), \quad \boldsymbol{\xi}\in\mathcal{T} \tag{20}$$

$$q(\boldsymbol{\xi}, t) = g(\boldsymbol{\xi}, t), \quad \text{if } \boldsymbol{n}\cdot\boldsymbol{V} < 0 \text{ for } \boldsymbol{\xi}\in\gamma\subset\partial\mathcal{T}, \quad t\geq 0 \tag{21}$$

We now conduct an energy estimate of the problem. The governing equation can be expressed in the following split form:

$$\partial_t q = -\frac{1}{2}\nabla\cdot(\boldsymbol{V}q) - \frac{1}{2}\boldsymbol{V}\cdot\nabla q - \frac{1}{2}(\nabla\cdot\boldsymbol{V})q \tag{22}$$

Multiplying $2q$ to the equation, integrating over the domain $\mathcal{T}$, and invoking the identity $\nabla\cdot(\boldsymbol{V}q^2) = q\nabla\cdot(\boldsymbol{V}q) + q\boldsymbol{V}\cdot\nabla q$ as well the divergence theorem, we have

$$\begin{aligned}\frac{d}{dt}\int_{\mathcal{T}} q^2(\boldsymbol{\xi},t)\,d\boldsymbol{\xi} &= -\int_{\mathcal{T}} q\nabla\cdot(\boldsymbol{V}q) + q\boldsymbol{V}\cdot\nabla q + (\nabla\cdot\boldsymbol{V})q^2\,d\boldsymbol{\xi}\\ &= -\int \nabla\cdot(\boldsymbol{V}q^2) + (\nabla\cdot\boldsymbol{V})q^2\,d\boldsymbol{\xi}\\ &= -\oint_{\partial\mathcal{T}}(n\cdot\boldsymbol{V})q^2 d\boldsymbol{\xi} - \int_{\mathcal{T}}(\nabla\cdot\boldsymbol{V})q^2\,d\boldsymbol{\xi}\end{aligned} \tag{23}$$

Let $\alpha = \max_{\boldsymbol{\xi}\in\mathcal{T}}|\nabla\cdot\boldsymbol{V}|$.

$$\frac{d}{dt}\int_{\mathcal{T}} q^2(\boldsymbol{\xi},t)\,d\boldsymbol{\xi} - \alpha\int_{\mathcal{T}} q^2\,d\boldsymbol{\xi} \le \int_{\gamma}(-n\cdot\boldsymbol{V})g^2\,d\boldsymbol{\xi} \tag{24}$$

Multiplying $e^{-\alpha t}$ to the equation and integrating with respect to t, we have

$$e^{-\alpha t}\int_{\mathcal{T}} q^2(\boldsymbol{\xi},t)\,d\boldsymbol{\xi} - \int_{\mathcal{T}} q^2(\boldsymbol{\xi},0)\,d\boldsymbol{\xi} \le \int_0^t e^{-\alpha t'}\int_{\gamma}(-n\cdot\boldsymbol{V})g^2\,d\boldsymbol{\xi}\,dt \tag{25}$$

leading to

$$\int_{\mathcal{T}} q^2(\boldsymbol{\xi},t)\,d\boldsymbol{\xi} \le e^{\alpha t}\int_{\mathcal{T}} q_0^2(\boldsymbol{\xi})\,d\boldsymbol{\xi} + \int_0^t e^{\alpha(t-t')}\int_{\gamma}(-n\cdot\boldsymbol{V})g^2(t')\,d\boldsymbol{\xi}\,dt' \tag{26}$$

3.5. DG scheme for advection equation

We now proceed constructing a DG scheme for solving advection equation.

We consider the following discretization:

$$\begin{aligned}\partial_t\boldsymbol{q} =& -\frac{1}{2}\left(\boldsymbol{D}_\xi(\boldsymbol{u}\odot\boldsymbol{q}) + \boldsymbol{D}_\eta(\boldsymbol{v}\odot\boldsymbol{q})\right) - \frac{1}{2}(\boldsymbol{u}\odot\boldsymbol{D}_\xi\boldsymbol{q} + \boldsymbol{v}\odot\boldsymbol{D}_\eta\boldsymbol{q}) - \frac{1}{2}(\boldsymbol{D}_\xi\boldsymbol{u} + \boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}\\ &+ \boldsymbol{V}\boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p}\end{aligned} \tag{27}$$

where

$$\boldsymbol{E} = \left[\boldsymbol{I}_{3(n+1)\times 3(n+1)}|\boldsymbol{0}_{3(n+1)\times(M-3(n+1))}\right] \tag{28}$$

and $\boldsymbol{p}$ is the penalized boundary condition vector, and it is form is to be determined later.

Multiplying $2|T|\boldsymbol{q}^T\boldsymbol{W}$ from the left to the scheme and invoking the discrete divergence theorem, we have

$$\begin{aligned}\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) =& -|T|\boldsymbol{q}^T\boldsymbol{W}\left(\boldsymbol{D}_\xi(\boldsymbol{u}\odot\boldsymbol{q}) + \boldsymbol{D}_\eta(\boldsymbol{v}\odot\boldsymbol{q})\right) - |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{u}\odot\boldsymbol{D}_\xi\boldsymbol{q} + \boldsymbol{v}\odot\boldsymbol{D}_\eta\boldsymbol{q})\\ &- |T|\boldsymbol{q}^T\boldsymbol{W}\left((\boldsymbol{D}_\xi\boldsymbol{u} + \boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}\right) + 2|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{V}\boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p}\end{aligned} \tag{29}$$

Apply the relationships $\boldsymbol{M}^{-1} = |T|^{-1}(\boldsymbol{V}\boldsymbol{W}\boldsymbol{V}^T)^{-1}$ and $\boldsymbol{q} = \boldsymbol{V}\hat{\boldsymbol{q}}$, the penalized term becomes

$$
\begin{aligned}
2|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{V}\boldsymbol{M}^{-1}\boldsymbol{V}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p} =& 2|T|(|T|)^{-1}\widehat{\boldsymbol{q}}^T\boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}(\boldsymbol{V}\boldsymbol{W}\boldsymbol{V}^T)^{-1}\boldsymbol{V}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p} \\
=& 2\widehat{\boldsymbol{q}}^T\boldsymbol{V}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p} = 2\boldsymbol{q}^T\boldsymbol{E}^T\boldsymbol{W}_b\boldsymbol{p} = 2\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{p} \qquad (30)
\end{aligned}
$$

Then, the energy rate becomes

$$
\begin{aligned}
\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) =& -\oint_{\partial\mathcal{T}}(\boldsymbol{n}\cdot\boldsymbol{V})q^2\,d\boldsymbol{x} + |T|(\boldsymbol{u}\odot\boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\xi\boldsymbol{q} + |T|(\boldsymbol{v}\odot\boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\eta\boldsymbol{q} \\
&- |T|\boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{u}\odot\boldsymbol{D}_\xi\boldsymbol{q} + \boldsymbol{v}\odot\boldsymbol{D}_\eta\boldsymbol{q}) - |T|\boldsymbol{q}^T\boldsymbol{W}\left((\boldsymbol{D}_\xi\boldsymbol{u} + \boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}\right) + 2\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{p} \qquad (31)
\end{aligned}
$$

Note that

$$
(\boldsymbol{u}\odot\boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\xi\boldsymbol{q} = \sum_{i=0}^{M} u_i q_i(\boldsymbol{D}_\xi\boldsymbol{q})_i w_i = \boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{u}\odot\boldsymbol{D}_\xi\boldsymbol{q}), \qquad (32)
$$

$$
(\boldsymbol{v}\odot\boldsymbol{q})^T\boldsymbol{W}\boldsymbol{D}_\eta\boldsymbol{q} = \sum_{i=0}^{M} v_i q_i(\boldsymbol{D}_\eta\boldsymbol{q})_i w_i = \boldsymbol{q}^T\boldsymbol{W}(\boldsymbol{v}\odot\boldsymbol{D}_\xi\boldsymbol{q}), \qquad (33)
$$

Hence,

$$
\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) = -\oint_{\partial\mathcal{T}}(\boldsymbol{n}\cdot\boldsymbol{V})q^2\,d\boldsymbol{x} - |T|\boldsymbol{q}^T\boldsymbol{W}((\boldsymbol{D}_\xi\boldsymbol{u} + \boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}) + 2\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{p} \qquad (34)
$$

Consider the boundary term as

$$
2\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{p} = \oint_{\partial\mathcal{T}} 2q\left(\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*)\right)\,d\boldsymbol{\xi} \qquad (35)
$$

where

$$
\boldsymbol{f}(q^-) = \boldsymbol{V}q^-, \quad \boldsymbol{f}^*(q^-, q^+) = \frac{\boldsymbol{V}q^- + \boldsymbol{V}q^+}{2} + (1-\tau)\frac{|\boldsymbol{n}^-\cdot\boldsymbol{V}|}{2}(\boldsymbol{n}^-q^- + \boldsymbol{n}^+q^+) \qquad (36)
$$

where $0 \le \tau \le 1$.

3.6. Numerical boundary and interface conditions

3.6.1. Boundary condition by upwind flux ($\tau = 0$, $q^- = q$ and $q^+ = g$)

To enforce boundary condition, we have, for $q^- = q$, $q^+ = g$, $\boldsymbol{n}^- = \boldsymbol{n}$, $\boldsymbol{n}^+ = -\boldsymbol{n}$,

$$
\boldsymbol{f}(q) = \boldsymbol{V}q, \quad \boldsymbol{f}^*(q, g) = \frac{\boldsymbol{V}q + \boldsymbol{V}g}{2} + \frac{|\boldsymbol{n}\cdot\boldsymbol{V}|}{2}(\boldsymbol{n}q - \boldsymbol{n}g)
$$

We now discuss the impact of the numerical flux f^*.

Case 1: out flow: $\boldsymbol{n}\cdot\boldsymbol{V} \ge 0$.

we have

$$
q\boldsymbol{n}\cdot\boldsymbol{f}^* = \frac{\boldsymbol{n}\cdot\boldsymbol{V}q^2 + \boldsymbol{n}\cdot\boldsymbol{V}qg}{2} + \frac{\boldsymbol{n}\cdot\boldsymbol{V}}{2}(q^2 - qg) = \boldsymbol{n}\cdot\boldsymbol{V}q^2, \qquad (37)
$$

$$
2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*) = 2q\boldsymbol{n}\cdot\boldsymbol{f} - 2q\boldsymbol{n}\cdot\boldsymbol{f}^* = 2(\boldsymbol{n}\cdot\boldsymbol{V})q^2 - 2(\boldsymbol{n}\cdot\boldsymbol{V})q^2 = 0 \qquad (38)
$$

Case 2: inflow: $\boldsymbol{n}\cdot\boldsymbol{V}<0$, $\boldsymbol{n}^-=\boldsymbol{n}$, $\boldsymbol{n}^+=-\boldsymbol{n}$.

$$q\boldsymbol{n}\cdot\boldsymbol{f}^* = \frac{\boldsymbol{n}\cdot\boldsymbol{V}q^2+\boldsymbol{n}\cdot\boldsymbol{V}qg}{2} - \frac{\boldsymbol{n}\cdot\boldsymbol{V}}{2}(q^2-qg) = \boldsymbol{n}\cdot\boldsymbol{V}qg$$

and

$$2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*) = 2q\boldsymbol{n}\cdot\boldsymbol{f} - 2q\boldsymbol{n}\cdot\boldsymbol{f}^* = 2(\boldsymbol{n}\cdot\boldsymbol{V})(q^2-qg)$$

Thus, the boundary term becomes

$$\oint_{\partial\mathcal{T}} \overbrace{(-\boldsymbol{n}\cdot\boldsymbol{V})q^2 + 2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*)}^{\rho}\, d\boldsymbol{x} \le \int_\gamma (-\boldsymbol{n}\cdot\boldsymbol{V})g^2\, d\boldsymbol{\xi} \tag{39}$$

since

$$\boldsymbol{\rho} = \begin{cases} (-\boldsymbol{n}\cdot\boldsymbol{V})q^2 \le 0 & \text{for out flowing } (\boldsymbol{n}\cdot\boldsymbol{V}>0) \\ (\boldsymbol{n}\cdot\boldsymbol{V})(q-g)^2 - \boldsymbol{n}\cdot\boldsymbol{V}g^2 \le (-\boldsymbol{n}\cdot\boldsymbol{V})g^2 & \text{for in flowing } (\boldsymbol{n}\cdot\boldsymbol{V}<0) \end{cases} \tag{40}$$

If $\max_{\boldsymbol{\xi}\in\mathcal{T}} ||-\boldsymbol{D}_\xi\boldsymbol{u}-\boldsymbol{D}_\eta\boldsymbol{v}||=\alpha$ then we have the energy rate as

$$|T|\frac{d}{dt}(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) - \alpha|T|(\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q}) \le -\oint_\gamma (\boldsymbol{n}\cdot\boldsymbol{V})g^2\, d\boldsymbol{\xi} \tag{41}$$

Multiplying the above expression with respect to time and applying initial condition, we have

$$(|T|\boldsymbol{q}^T\boldsymbol{W}\boldsymbol{q})(t) \le e^{\alpha t}|T|\boldsymbol{q}_0^T\boldsymbol{W}\boldsymbol{q}_0 + \int_0^t e^{\alpha(t-t')}\int_{\partial T}(-\boldsymbol{n}\cdot\boldsymbol{V})g^2(\boldsymbol{\xi},t')\, d\boldsymbol{\xi}\, dt' \tag{42}$$

We are left with the exact form of $\boldsymbol{p}$. Let

$$\boldsymbol{q} = [\overbrace{q_1,q_2,\ldots,q_{3(n+1)}}^{\text{boundary points}}, \overbrace{q_{3(n+1)+1},\ldots,q_M}^{\text{interior points}}]^T, \quad \boldsymbol{p} = [\overbrace{p_1,p_2,\ldots,p_{3(n+1)}}^{\text{boundary points}}]^T. \tag{43}$$

We require that

$$\begin{aligned} 2\boldsymbol{q}_b^T\boldsymbol{W}_b\boldsymbol{p} &= \sum_{i=1}^{n+1} 2|\partial T^1|q_i w_i^b p_i + \sum_{i=1}^{n+1} 2|\partial T^2|q_{n+1+i}w_i^b p_{n+1+i} + \sum_{i=1}^{n+1} 2|\partial T^2|q_{2(n+1)+i}w_i^b p_{2(n+1)+i} \\ &= \oint_{\partial T} 2q\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*)\, d\boldsymbol{x} \\ &= \sum_{j=0}^{2}\sum_{i=1}^{n+1} 2|\partial T^{j+1}|\, q_{j(n+1)+i}\, (\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_{j(n+1)+i} w_i^b \end{aligned} \tag{44}$$

Thus,

$$p_i = (\boldsymbol{n}\cdot(\boldsymbol{f}-\boldsymbol{f}^*))\Big|_i \tag{45}$$

where $f_i = f(\boldsymbol{\xi}_i)$ and $f_i^* = f^*(\boldsymbol{\xi}_i)$.

3.6.2. interface condition by central flux ($\tau = 1$, $q^- = q$ and $q^+ = g$)

The energy rate equations are

$$\forall k = 1, 2, \quad \frac{d}{dt}(|T|(\boldsymbol{q}^{(k)})^T \boldsymbol{W} \boldsymbol{q}^{(k)}) = -\oint_{\partial\mathcal{T}} (\boldsymbol{n}^{(k)} \cdot \boldsymbol{V})(q^{(k)})^2 \, d\boldsymbol{x} + 2(\boldsymbol{q}_b^{(k)})^T \boldsymbol{W}_b \boldsymbol{p}^{(k)} - |T|(\boldsymbol{q}^{(k)})^T \boldsymbol{W}((\boldsymbol{D}_\xi \boldsymbol{u} + \boldsymbol{D}_\eta \boldsymbol{v}) \odot \boldsymbol{q}^{(k)}) \tag{46}$$

To enforce interface condition, we introduce the following flux and numerical flux functions:

$$\boldsymbol{f}(\boldsymbol{q}^-) = \boldsymbol{V}\boldsymbol{q}^-, \quad \boldsymbol{f}^*(\boldsymbol{q}^-, \boldsymbol{q}^+) = \frac{\boldsymbol{V}q^- + \boldsymbol{V}q^+}{2} + \frac{C}{2}(\boldsymbol{n}^- q^- + \boldsymbol{n}^+ q^+) \tag{47}$$

Note that if $C = 0$ then $\boldsymbol{f}^*$ is the central flux function. If $\boldsymbol{n}^- = \boldsymbol{n} = -\boldsymbol{n}^+$ and $C = |\boldsymbol{n} \cdot \boldsymbol{V}|$ then we recover the upwind flux function.

To enforce interface condition from the domain (1) side, we have, for $q^- = q^{(1)}$, $q^+ = q^{(2)}$, $\boldsymbol{n}^- = \boldsymbol{n}^{(1)}$, $\boldsymbol{n}^+ = \boldsymbol{n}^{(2)} = -\boldsymbol{n}^{(1)}$,

$$\boldsymbol{f}(q^{(1)}) = \boldsymbol{V}q^{(1)}, \quad \boldsymbol{f}^*(q^{(1)}, q^{(2)}) = \frac{\boldsymbol{V}q^{(1)} + \boldsymbol{V}q^{(2)}}{2} + \frac{C}{2}\boldsymbol{n}^{(1)}(q^{(1)} - q^{(2)})$$

and to enforce interface condition from the domain (2) side, we have, for $q^- = q^{(2)}$, $q^+ = q^{(1)}$, $\boldsymbol{n}^- = \boldsymbol{n}^{(2)}$, $\boldsymbol{n}^+ = \boldsymbol{n}^{(1)} = -\boldsymbol{n}^{(2)}$,

$$\boldsymbol{f}(q^{(2)}) = \boldsymbol{V}q^{(2)}, \quad \boldsymbol{f}^*(q^{(2)}, q^{(1)}) = \frac{\boldsymbol{V}q^{(2)} + \boldsymbol{V}q^{(1)}}{2} + \frac{C}{2}\boldsymbol{n}^{(2)}(q^{(2)} - q^{(1)})$$

For domain (1), we have

$$\begin{aligned}
2(\boldsymbol{q}_b^{(1)})^T \boldsymbol{W}_b \boldsymbol{p}^{(1)} &= \oint_{\partial\mathcal{T}} 2q^{(1)} \left(\boldsymbol{n}^{(1)} \cdot (\boldsymbol{f}(q^{(1)}) - \boldsymbol{f}^*(q^{(1)}, q^{(2)})) \right) d\boldsymbol{\xi} \\
&= \oint_{\partial\mathcal{T}} 2 \left(\boldsymbol{n}^{(1)} \cdot \boldsymbol{V}(q^{(1)})^2 - \boldsymbol{n}^{(1)} \cdot \boldsymbol{V}\frac{(q^{(1)})^2 + q^{(1)}q^{(2)}}{2} - \frac{C}{2}((q^{(1)})^2 - q^{(1)}q^{(2)}) \right) d\boldsymbol{\xi} \\
&= \oint_{\partial\mathcal{T}} \left(\boldsymbol{n}^{(1)} \cdot \boldsymbol{V} \right) (q^{(1)})^2 \, d\boldsymbol{\xi} - \oint_{\partial\mathcal{T}} \left(\boldsymbol{n}^{(1)} \cdot \boldsymbol{V} \right) q^{(1)}q^{(2)} \, d\boldsymbol{\xi} \\
&\quad - \oint_{\partial\mathcal{T}} C \left((q^{(1)})^2 - q^{(1)}q^{(2)} \right) d\boldsymbol{\xi}
\end{aligned} \tag{48}$$

For domain (2), we have

$$\begin{aligned}
2(\boldsymbol{q}_b^{(2)})^T \boldsymbol{W}^b \boldsymbol{p}^{(2)} &= \oint_{\partial\mathcal{T}} 2q^{(2)} \left(\boldsymbol{n}^{(2)} \cdot \left(\boldsymbol{f}(q^{(2)}) - \boldsymbol{f}^*(q^{(2)}, q^{(1)}) \right) \right) d\boldsymbol{\xi} \\
&= \oint_{\partial\mathcal{T}} 2 \left(\boldsymbol{n}^{(2)} \cdot \boldsymbol{V}(q^{(2)})^2 - \boldsymbol{n}^{(2)} \cdot \boldsymbol{V}\frac{(q^{(2)})^2 + q^{(2)}q^{(1)}}{2} - \frac{C}{2} \left((q^{(2)})^2 - q^{(2)}q^{(1)} \right) \right) d\boldsymbol{\xi} \\
&= \oint_{\partial\mathcal{T}} \left(\boldsymbol{n}^{(2)} \cdot \boldsymbol{V} \right) (q^{(2)})^2 \, d\boldsymbol{\xi} - \oint_{\partial\mathcal{T}} \left(\boldsymbol{n}^{(2)} \cdot \boldsymbol{V} \right) q^{(2)}q^{(1)} \, d\boldsymbol{\xi} \\
&\quad - \oint_{\partial\mathcal{T}} C \left((q^{(2)})^2 - q^{(2)}q^{(1)} \right) d\boldsymbol{\xi}
\end{aligned} \tag{49}$$

The total energy rate equation s

$$\frac{d}{dt}\sum_{k=1}^{2}(|T|(\boldsymbol{q}^{(k)})^T\boldsymbol{W}\boldsymbol{q}^{(k)}) = -\sum_{k=1}^{2}\oint_{\partial\mathcal{T}}(\boldsymbol{n}^{(k)}\cdot\boldsymbol{V})(q^{(k)})^2\,d\boldsymbol{x} + \sum_{k=1}^{2}2(\boldsymbol{q}_b^{(k)})^T\boldsymbol{W}_b\boldsymbol{p}^{(k)}$$
$$-\sum_{k=1}^{2}|T|(\boldsymbol{q}^{(k)})^T\boldsymbol{W}((\boldsymbol{D}_\xi\boldsymbol{u}+\boldsymbol{D}_\eta\boldsymbol{v})\odot\boldsymbol{q}^{(k)}) \tag{50}$$

Thus, at an interface point the energy is

$$-C\left((q^{(1)})^2 - q^{(1)}q^{(2)}\right) - C\left((q^{(2)})^2 - q^{(2)}q^{(1)}\right) = -C\left(q^{(1)} - q^{(2)}\right)^2$$

Thus, to ensure dissipation at the interface, we may choose C as follows,

$$C = \begin{cases} |\boldsymbol{n}\cdot\boldsymbol{V}| & \text{upwind flux} \\ \alpha\max_T|\boldsymbol{V}|, \quad \alpha\geq 1 & \text{Lax-Friedrichs flux} \end{cases} \tag{51}$$

We shown that if $C > 0$ the scheme is stable in the semi-discrete level. However, at the fully-discrete level, the scheme may not be stable, due to the amount of dissipation is not enough. Hence, we may adjust the value of α so that the scheme can be made stable. However, the larger the value of α, the smaller time step is required.

Octahedron Sphere

We now discuss the mapping from a box $(x, y) \in [-1, 1]^2$ to a octahedron sphere.

3.7. T_1 to O_1

Let T_1 (triangle), O_1 (octahedron) and S_1 (sphere) be defined as

$$\begin{aligned} T_1 &= \{(x,y)\,|\,x+y\leq 1, \quad 0\leq x,y\leq 1\}, \\ O_1 &= \{(X,Y,Z)\,|\,X+Y+Z=R, \quad 0\leq X,Y,Z\leq R\} \\ S_1 &= \{(\rho=R,\lambda,\theta)\,|\,0\leq\lambda\leq\frac{\pi}{2}, \quad 0\leq\theta\leq\frac{\pi}{2}\} \\ (X,Y,Z) &= R(x,y,1-x-y). \end{aligned}$$

Then the mapping from (x, y) to (λ, θ) is as follows,

$$\lambda = \frac{\pi}{2}\frac{y}{x+y}, \quad \sin\theta = 1-(x+y)^2, \quad (x,y)\in T_1,$$

or

$$x = \left(1-\frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}, \quad y = \frac{2\lambda}{\pi}\sqrt{1-\sin\theta}$$

Then

$$(X,Y,Z)=R(x,y,1-x-y),\quad (\bar{X},\bar{Y},\bar{Z})=R(\cos\lambda\cos\theta,\sin\lambda\cos\theta,\sin\theta)$$

Compute

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)}=\begin{bmatrix}x_\lambda & x_\theta\\ y_\lambda & y_\theta\end{bmatrix}=\begin{bmatrix}-\frac{2}{\pi}\sqrt{1-\sin\theta} & -\frac{(\pi-2\lambda)\cos\theta}{2\pi\sqrt{1-\sin\theta}}\\ \frac{2}{\pi}\sqrt{1-\sin\theta} & -\frac{2\lambda\cos\theta}{2\pi\sqrt{1-\sin\theta}}\end{bmatrix}=\frac{1}{2\pi\sqrt{1-\sin\theta}}\begin{bmatrix}-4(1-\sin\theta) & -(\pi-2\lambda)\cos\theta\\ 4(1-\sin\theta) & -2\lambda\cos\theta\end{bmatrix}$$

and

$$\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right)=\frac{\cos\theta}{\pi}$$

Then

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)}=\begin{bmatrix}\lambda_x & \lambda_y\\ \theta_x & \theta_y\end{bmatrix}=\frac{1}{\cos\theta}\frac{1}{2\sqrt{1-\sin\theta}}\begin{bmatrix}-2\lambda\cos\theta & (\pi-2\lambda)\cos\theta\\ -4(1-\sin\theta) & -4(1-\sin\theta)\end{bmatrix}$$

Transformation matrix $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$:

$$\boldsymbol{A}=\begin{bmatrix}R\cos\theta\lambda_x & R\cos\theta\lambda_y\\ R\theta_x & R\theta_y\end{bmatrix}=\frac{R}{2\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix}-2\lambda\cos^2\theta & (\pi-2\lambda)\cos^2\theta\\ -4(1-\sin\theta) & -4(1-\sin\theta)\end{bmatrix}\tag{52}$$

$$\det(\boldsymbol{A})=\pi R^2\tag{53}$$

Thus,

$$\boldsymbol{A}^{-1}=\frac{1}{\det(\boldsymbol{A})}\begin{bmatrix}R\theta_y & -R\cos\theta\lambda_y\\ -R\theta_x & R\cos\theta\lambda_x\end{bmatrix}=\frac{1}{2\pi R}\frac{1}{\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix}-4(1-\sin\theta) & -(\pi-2\lambda)\cos^2\theta\\ 4(1-\sin\theta) & -2\lambda\cos^2\theta\end{bmatrix}\tag{54}$$

Note that $\theta\in[\pi,\frac{\pi}{2})$. Hence, as $\theta\to(\frac{\pi}{2})^-$

$$\boldsymbol{A}^{-1}=\frac{1}{2\pi R}\begin{bmatrix}-4\frac{\sqrt{1-\sin\theta}}{\cos\theta} & -(\pi-2\lambda)\frac{\cos\theta}{\sqrt{1+\sin\theta}}\\ 4\frac{\sqrt{1-\sin\theta}}{\cos\theta} & -2\lambda\frac{\cos\theta}{\sqrt{1-\sin\theta}}\end{bmatrix}=\frac{1}{\pi R}\begin{bmatrix}-\frac{2}{\sqrt{1+\sin\theta}} & -(\frac{\pi}{2}-\lambda)\sqrt{1+\sin\theta}\\ \frac{2}{\sqrt{1+\sin\theta}} & -\lambda\sqrt{1+\sin\theta}\end{bmatrix}\tag{55}$$

$\boldsymbol{G}$ and $\boldsymbol{G}^{-1}$ matrices and $\sqrt{G}$:

$$\begin{aligned}\boldsymbol{G}=&\boldsymbol{A}^T\boldsymbol{A}=\frac{R^2}{4\cos^2\theta(1-\sin\theta)}\begin{bmatrix}-2\lambda\cos^2\theta & -4(1-\sin\theta)\\ (\pi-2\lambda)\cos^2\theta & -4(1-\sin\theta)\end{bmatrix}\begin{bmatrix}-2\lambda\cos^2\theta & (\pi-2\lambda)\cos^2\theta\\ -4(1-\sin\theta) & -4(1-\sin\theta)\end{bmatrix}\\ =&\frac{R^2}{4\cos^2\theta(1-\sin\theta)}\begin{bmatrix}4\lambda^2\cos^4\theta+16(1-\sin\theta)^2 & -2\lambda(\pi-2\lambda)\cos^4\theta+16(1-\sin\theta)^2\\ -2\lambda(\pi-2\lambda)\cos^4\theta+16(1-\sin\theta)^2 & (\pi-2\lambda)^2\cos^4\theta+16(1-\sin\theta)^2\end{bmatrix}\end{aligned}$$

$$\sqrt{G}=\sqrt{\det(\boldsymbol{G})}=\pi R^2$$

Transformation of vector on the sphere: Let

$$\boldsymbol{V} = (u, v) = u\boldsymbol{e}_\lambda + v\boldsymbol{e}_\theta$$

Then

$$\begin{bmatrix} u \\ v \end{bmatrix} = \boldsymbol{A} \begin{bmatrix} u^1 \\ u^2 \end{bmatrix}, \quad \begin{bmatrix} u^1 \\ u^2 \end{bmatrix} = \boldsymbol{A}^{-1} \begin{bmatrix} u \\ v \end{bmatrix}$$

3.8. T_2 to O_2

Let

$$x = 1 - \frac{2\lambda}{\pi}\sqrt{1+\sin\theta}, \quad y = 1 - \left(1 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta}$$

Then

$$\begin{aligned} \theta &= \arcsin\left((2-x-y)^2 - 1\right) \\ \lambda &= \frac{\pi(1-x)}{2(2-x-y)} \end{aligned}$$

1. Jacobian Matrix $\frac{\partial(x,y)}{\partial(\lambda,\theta)}$

The partial derivatives are:

$$\begin{aligned} x_\lambda &= -\frac{2}{\pi}\sqrt{1+\sin\theta} \\ x_\theta &= -\frac{2\lambda}{\pi}\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = -\frac{\lambda\cos\theta}{\pi\sqrt{1+\sin\theta}} \\ y_\lambda &= \frac{2}{\pi}\sqrt{1+\sin\theta} \\ y_\theta &= -\left(1 - \frac{2\lambda}{\pi}\right)\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = -\frac{(\pi-2\lambda)\cos\theta}{2\pi\sqrt{1+\sin\theta}} \end{aligned}$$

The Jacobian matrix is:

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1+\sin\theta}} \begin{bmatrix} -4(1+\sin\theta) & -2\lambda\cos\theta \\ 4(1+\sin\theta) & -(\pi-2\lambda)\cos\theta \end{bmatrix}$$

2. Determinant $\det(J)$

$$\begin{aligned} \det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) &= \frac{1}{4\pi^2(1+\sin\theta)}\left[4(1+\sin\theta)(\pi-2\lambda)\cos\theta + 8\lambda(1+\sin\theta)\cos\theta\right] \\ &= \frac{4(1+\sin\theta)\cos\theta(\pi-2\lambda+2\lambda)}{4\pi^2(1+\sin\theta)} = \frac{\cos\theta}{\pi} \end{aligned}$$

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \begin{bmatrix} \lambda_x & \lambda_y \\ \theta_x & \theta_y \end{bmatrix} = \frac{1}{2\cos\theta\sqrt{1+\sin\theta}} \begin{bmatrix} -(\pi-2\lambda)\cos\theta & 2\lambda\cos\theta \\ -4(1+\sin\theta) & -4(1+\sin\theta) \end{bmatrix}$$

3. Transformation Matrix $\mathbf{A}$

Using $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$ and $J^{-1} = \frac{\pi}{\cos\theta}\mathrm{adj}(J)$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1+\sin\theta}} \begin{bmatrix} -(\pi-2\lambda)\cos^2\theta & 2\lambda\cos^2\theta \\ -4(1+\sin\theta) & -4(1+\sin\theta) \end{bmatrix}$$

The determinant is:

$$\det(\mathbf{A}) = R^2\cos\theta\cdot\frac{\pi}{\cos\theta} = \pi R^2$$

4. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$. *(*$S = \sin\theta$*)*

$$\begin{aligned} \mathbf{G} &= \frac{R^2}{4\cos^2\theta(1+\sin\theta)} \begin{bmatrix} -(\pi-2\lambda)\cos^2\theta & -4(1+S) \\ 2\lambda\cos^2\theta & -4(1+S) \end{bmatrix} \begin{bmatrix} -(\pi-2\lambda)\cos^2\theta & 2\lambda\cos^2\theta \\ -4(1+S) & -4(1+S) \end{bmatrix} \\ &= \frac{R^2}{4\cos^2\theta(1+\sin\theta)} \begin{bmatrix} (\pi-2\lambda)^2\cos^4\theta+16(1+\sin\theta)^2 & -2\lambda(\pi-2\lambda)\cos^4\theta+16(1+\sin\theta)^2 \\ -2\lambda(\pi-2\lambda)\cos^4\theta+16(1+\sin\theta)^2 & 4\lambda^2\cos^4\theta+16(1+\sin\theta)^2 \end{bmatrix} \end{aligned}$$

5. Resultant $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

3.9. T_3 *to* O_3

Let

$$\lambda = \pi - \frac{\pi}{2}\frac{y}{-x+y}, \quad \sin\theta = 1-(-x+y)^2$$

$$\begin{aligned} x &= \left(1-\frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta} \\ y &= \left(2-\frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta} \end{aligned}$$

Derivation for $x = \left(1 - \frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}$ and $y = \left(2 - \frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}$

1. Jacobian Matrix of (x, y) with respect to (λ, θ)

The partial derivatives are calculated as follows:

$$\begin{aligned}
\frac{\partial x}{\partial \lambda} &= -\frac{2}{\pi}\sqrt{1-\sin\theta} \\
\frac{\partial x}{\partial \theta} &= -\frac{(1-\frac{2\lambda}{\pi})\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{(\pi-2\lambda)\cos\theta}{2\pi\sqrt{1-\sin\theta}} \\
\frac{\partial y}{\partial \lambda} &= -\frac{2}{\pi}\sqrt{1-\sin\theta} \\
\frac{\partial y}{\partial \theta} &= -\frac{(2-\frac{2\lambda}{\pi})\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{(2\pi-2\lambda)\cos\theta}{2\pi\sqrt{1-\sin\theta}}
\end{aligned}$$

The Jacobian matrix is:

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1-\sin\theta}} \begin{bmatrix} -4(1-\sin\theta) & -(\pi-2\lambda)\cos\theta \\ -4(1-\sin\theta) & -(2\pi-2\lambda)\cos\theta \end{bmatrix}$$

2. Jacobian Determinant

$$\begin{aligned}
\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) &= \frac{1}{4\pi^2(1-\sin\theta)}\left[4(1-\sin\theta)(2\pi-2\lambda)\cos\theta - 4(1-\sin\theta)(\pi-2\lambda)\cos\theta\right] \\
&= \frac{4(1-\sin\theta)\cos\theta(2\pi-2\lambda-\pi+2\lambda)}{4\pi^2(1-\sin\theta)} = \frac{\cos\theta}{\pi}
\end{aligned}$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using the inverse formula $J^{-1} = \frac{1}{\det(J)}\operatorname{adj}(J)$:

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1-\sin\theta}} \begin{bmatrix} -(2\pi-2\lambda)\cos\theta & (\pi-2\lambda)\cos\theta \\ 4(1-\sin\theta) & -4(1-\sin\theta) \end{bmatrix}$$

4. Transformation Matrix $\mathbf{A}$

Defined by $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1-\sin\theta}} \begin{bmatrix} -(2\pi-2\lambda)\cos^2\theta & (\pi-2\lambda)\cos^2\theta \\ 4(1-\sin\theta) & -4(1-\sin\theta) \end{bmatrix}$$

The determinant is:

$$\det(\mathbf{A}) = R^2\cos\theta\,(\lambda_x\theta_y - \lambda_y\theta_x) = R^2\cos\theta \cdot \frac{\pi}{\cos\theta} = \pi R^2$$

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T \mathbf{A}$

$$\mathbf{G} = \frac{R^2}{4\cos^2\theta(1-\sin\theta)} \begin{bmatrix} (2\pi-2\lambda)^2\cos^4\theta + 16(1-\sin\theta)^2 & -2(\pi-\lambda)(\pi-2\lambda)\cos^4\theta - 16(1-\sin\theta)^2 \\ -2(\pi-\lambda)(\pi-2\lambda)\cos^4\theta - 16(1-\sin\theta)^2 & (\pi-2\lambda)^2\cos^4\theta + 16(1-\sin\theta)^2 \end{bmatrix}$$

6. Area Element $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

3.10. T_4 *to* O_4

Let

$$\sin\theta = (2+x-y)^2 - 1$$
$$\lambda = \pi - \frac{\pi(1+x)}{2(2+x-y)}$$

Then

$$x = \left(2 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta} - 1$$
$$y = 1 + \left(1 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta}$$

Derivation for $\boldsymbol{x = \left(2 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta} - 1}$ **and** $\boldsymbol{y = 1 + \left(1 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta}}$

1. Jacobian Matrix of (x, y) *with respect to* (λ, θ)

The partial derivatives are:

$$\frac{\partial x}{\partial \lambda} = -\frac{2}{\pi}\sqrt{1+\sin\theta}$$
$$\frac{\partial x}{\partial \theta} = \left(2 - \frac{2\lambda}{\pi}\right)\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = \frac{(2\pi-2\lambda)\cos\theta}{2\pi\sqrt{1+\sin\theta}}$$
$$\frac{\partial y}{\partial \lambda} = -\frac{2}{\pi}\sqrt{1+\sin\theta}$$
$$\frac{\partial y}{\partial \theta} = \left(1 - \frac{2\lambda}{\pi}\right)\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = \frac{(\pi-2\lambda)\cos\theta}{2\pi\sqrt{1+\sin\theta}}$$

The Jacobian matrix is:

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1+\sin\theta}} \begin{bmatrix} -4(1+\sin\theta) & (2\pi-2\lambda)\cos\theta \\ -4(1+\sin\theta) & (\pi-2\lambda)\cos\theta \end{bmatrix}$$

2. Jacobian Determinant

$$\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) = \frac{1}{4\pi^2(1+\sin\theta)}\left[-4(1+\sin\theta)(\pi-2\lambda)\cos\theta + 4(1+\sin\theta)(2\pi-2\lambda)\cos\theta\right]$$
$$= \frac{4(1+\sin\theta)\cos\theta(-\pi+2\lambda+2\pi-2\lambda)}{4\pi^2(1+\sin\theta)} = \frac{\cos\theta}{\pi}$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using $J^{-1} = \frac{\pi}{\cos\theta}\mathrm{adj}(J)$:

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1+\sin\theta}}\begin{bmatrix} (\pi-2\lambda)\cos\theta & -(2\pi-2\lambda)\cos\theta \\ 4(1+\sin\theta) & -4(1+\sin\theta) \end{bmatrix}$$

4. Transformation Matrix $\mathbf{A}$

Based on $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1+\sin\theta}}\begin{bmatrix} (\pi-2\lambda)\cos^2\theta & -(2\pi-2\lambda)\cos^2\theta \\ 4(1+\sin\theta) & -4(1+\sin\theta) \end{bmatrix}$$

The determinant remains $\det(\mathbf{A}) = \pi R^2$.

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$

$$\mathbf{G} = \frac{R^2}{4\cos^2\theta(1+\sin\theta)}\begin{bmatrix} (\pi-2\lambda)^2\cos^4\theta + 16(1+\sin\theta)^2 & -(\pi-2\lambda)(2\pi-2\lambda)\cos^4\theta - 16(1+\sin\theta)^2 \\ -(\pi-2\lambda)(2\pi-2\lambda)\cos^4\theta - 16(1+\sin\theta)^2 & (2\pi-2\lambda)^2\cos^4\theta + 16(1+\sin\theta)^2 \end{bmatrix}$$

6. Square Root of Determinant $\sqrt{G}$

$$\sqrt{G} = \pi R^2$$

3.11. T_5 to O_5

Let

$$\sin\theta = 1-(x+y)^2$$
$$\lambda = \pi + \frac{\pi y}{2(x+y)}$$

Then

$$x = \left(3-\frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}$$
$$y = \left(\frac{2\lambda}{\pi}-2\right)\sqrt{1-\sin\theta}$$

Derivation for $\boldsymbol{x = \left(3 - \frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}}$ and $\boldsymbol{y = \left(\frac{2\lambda}{\pi} - 2\right)\sqrt{1-\sin\theta}}$

1. Jacobian Matrix of (x, y) with respect to (λ, θ)

The partial derivatives are:

$$\begin{aligned}
x_\lambda &= -\frac{2}{\pi}\sqrt{1-\sin\theta} \\
x_\theta &= \left(3 - \frac{2\lambda}{\pi}\right)\frac{-\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{(3\pi - 2\lambda)\cos\theta}{2\pi\sqrt{1-\sin\theta}} \\
y_\lambda &= \frac{2}{\pi}\sqrt{1-\sin\theta} \\
y_\theta &= \left(\frac{2\lambda}{\pi} - 2\right)\frac{-\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{(2\lambda - 2\pi)\cos\theta}{2\pi\sqrt{1-\sin\theta}}
\end{aligned}$$

The Jacobian matrix is:

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1-\sin\theta}}\begin{bmatrix} -4(1-\sin\theta) & -(3\pi - 2\lambda)\cos\theta \\ 4(1-\sin\theta) & -(2\lambda - 2\pi)\cos\theta \end{bmatrix}$$

2. Jacobian Determinant

$$\begin{aligned}
\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) &= \frac{1}{4\pi^2(1-\sin\theta)}\left[4(1-\sin\theta)(2\lambda - 2\pi)\cos\theta + 4(1-\sin\theta)(3\pi - 2\lambda)\cos\theta\right] \\
&= \frac{4(1-\sin\theta)\cos\theta(2\lambda - 2\pi + 3\pi - 2\lambda)}{4\pi^2(1-\sin\theta)} = \frac{\pi\cos\theta}{\pi^2} = \frac{\cos\theta}{\pi}
\end{aligned}$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using $J^{-1} = \frac{\pi}{\cos\theta}\operatorname{adj}(J)$:

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix} -(2\lambda - 2\pi)\cos\theta & (3\pi - 2\lambda)\cos\theta \\ -4(1-\sin\theta) & -4(1-\sin\theta) \end{bmatrix}$$

4. Transformation Matrix $\mathbf{A}$

Based on $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix} -(2\lambda - 2\pi)\cos^2\theta & (3\pi - 2\lambda)\cos^2\theta \\ -4(1-\sin\theta) & -4(1-\sin\theta) \end{bmatrix}$$

The determinant is $\det(\mathbf{A}) = \pi R^2$.

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$

$$\mathbf{G} = \frac{R^2}{4\cos^2\theta(1-\sin\theta)}\begin{bmatrix} (2\lambda - 2\pi)^2\cos^4\theta + 16(1-\sin\theta)^2 & -(2\lambda - 2\pi)(3\pi - 2\lambda)\cos^4\theta + 16(1-\sin\theta)^2 \\ -(2\lambda - 2\pi)(3\pi - 2\lambda)\cos^4\theta + 16(1-\sin\theta)^2 & (3\pi - 2\lambda)^2\cos^4\theta + 16(1-\sin\theta)^2 \end{bmatrix}$$

6. Area Element $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

T_6 to O_6

Let

$$\begin{aligned}
\sin\theta &= (2 + x + y)^2 - 1 \\
\lambda &= \pi + \frac{\pi(1+x)}{2(2+x+y)}
\end{aligned}$$

Then

$$\begin{aligned}
x &= \left(\frac{2\lambda}{\pi} - 2\right)\sqrt{1+\sin\theta} - 1 \\
y &= \left(3 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta} - 1
\end{aligned}$$

Derivation for $x = \left(\frac{2\lambda}{\pi} - 2\right)\sqrt{1+\sin\theta} - 1$ and $y = \left(3 - \frac{2\lambda}{\pi}\right)\sqrt{1+\sin\theta} - 1$

1. Jacobian Matrix of (x, y) with respect to (λ, θ)

The partial derivatives are:

$$\begin{aligned}
x_\lambda &= \frac{2}{\pi}\sqrt{1+\sin\theta} \\
x_\theta &= \left(\frac{2\lambda}{\pi} - 2\right)\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = \frac{(2\lambda - 2\pi)\cos\theta}{2\pi\sqrt{1+\sin\theta}} \\
y_\lambda &= -\frac{2}{\pi}\sqrt{1+\sin\theta} \\
y_\theta &= \left(3 - \frac{2\lambda}{\pi}\right)\frac{\cos\theta}{2\sqrt{1+\sin\theta}} = \frac{(3\pi - 2\lambda)\cos\theta}{2\pi\sqrt{1+\sin\theta}}
\end{aligned}$$

The Jacobian matrix is:

$$\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1+\sin\theta}}\begin{bmatrix} 4(1+\sin\theta) & (2\lambda - 2\pi)\cos\theta \\ -4(1+\sin\theta) & (3\pi - 2\lambda)\cos\theta \end{bmatrix}$$

2. Jacobian Determinant

$$\begin{aligned}
\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) &= \frac{1}{4\pi^2(1+\sin\theta)}\left[4(1+\sin\theta)(3\pi - 2\lambda)\cos\theta + 4(1+\sin\theta)(2\lambda - 2\pi)\cos\theta\right] \\
&= \frac{4(1+\sin\theta)\cos\theta(3\pi - 2\lambda + 2\lambda - 2\pi)}{4\pi^2(1+\sin\theta)} = \frac{\pi\cos\theta}{\pi^2} = \frac{\cos\theta}{\pi}
\end{aligned}$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using $J^{-1} = \frac{\pi}{\cos\theta}\mathrm{adj}(J)$:

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1+\sin\theta}} \begin{bmatrix} (3\pi - 2\lambda)\cos\theta & -(2\lambda - 2\pi)\cos\theta \\ 4(1+\sin\theta) & 4(1+\sin\theta) \end{bmatrix}$$

4. Transformation Matrix $\mathbf{A}$

Based on $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1+\sin\theta}} \begin{bmatrix} (3\pi - 2\lambda)\cos^2\theta & -(2\lambda - 2\pi)\cos^2\theta \\ 4(1+\sin\theta) & 4(1+\sin\theta) \end{bmatrix}$$

The determinant is $\det(\mathbf{A}) = \pi R^2$.

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$

$$\mathbf{G} = \frac{R^2}{4\cos^2\theta(1+\sin\theta)} \begin{bmatrix} (3\pi-2\lambda)^2\cos^4\theta + 16(1+\sin\theta)^2 & -(3\pi-2\lambda)(2\lambda-2\pi)\cos^4\theta + 16(1+\sin\theta)^2 \\ -(3\pi-2\lambda)(2\lambda-2\pi)\cos^4\theta + 16(1+\sin\theta)^2 & (2\lambda-2\pi)^2\cos^4\theta + 16(1+\sin\theta)^2 \end{bmatrix}$$

6. Area Element $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

3.12. T_7 *to* O_7

Let

$$\lambda = -\frac{\pi}{2}\left(\frac{-y}{x-y}\right), \quad \sin\theta = 1 - (x-y)^2$$

Then

$$x = \left(1 + \frac{2\lambda}{\pi}\right)\sqrt{1-\sin\theta}$$
$$y = \frac{2\lambda}{\pi}\sqrt{1-\sin\theta}$$

Derivation for $x = \left(1 + \frac{2\lambda}{\pi}\right)\sqrt{1 - \sin\theta}$ and $y = \frac{2\lambda}{\pi}\sqrt{1 - \sin\theta}$

1. Jacobian Matrix of (x, y) with respect to (λ, θ)

The partial derivatives are:

$$
\begin{aligned}
x_\lambda &= \frac{2}{\pi}\sqrt{1-\sin\theta} \\
x_\theta &= \left(1 + \frac{2\lambda}{\pi}\right)\frac{-\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{(\pi + 2\lambda)\cos\theta}{2\pi\sqrt{1-\sin\theta}} \\
y_\lambda &= \frac{2}{\pi}\sqrt{1-\sin\theta} \\
y_\theta &= \frac{2\lambda}{\pi}\frac{-\cos\theta}{2\sqrt{1-\sin\theta}} = -\frac{2\lambda\cos\theta}{2\pi\sqrt{1-\sin\theta}}
\end{aligned}
$$

The Jacobian matrix is:

$$
\frac{\partial(x,y)}{\partial(\lambda,\theta)} = \frac{1}{2\pi\sqrt{1-\sin\theta}}\begin{bmatrix} 4(1-\sin\theta) & -(\pi+2\lambda)\cos\theta \\ 4(1-\sin\theta) & -2\lambda\cos\theta \end{bmatrix}
$$

2. Jacobian Determinant

$$
\begin{aligned}
\det\left(\frac{\partial(x,y)}{\partial(\lambda,\theta)}\right) &= \frac{1}{4\pi^2(1-\sin\theta)}\left[-8\lambda(1-\sin\theta)\cos\theta + 4(1-\sin\theta)(\pi+2\lambda)\cos\theta\right] \\
&= \frac{4(1-\sin\theta)\cos\theta(-2\lambda+\pi+2\lambda)}{4\pi^2(1-\sin\theta)} = \frac{\cos\theta}{\pi}
\end{aligned}
$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using $J^{-1} = \frac{\pi}{\cos\theta}\mathrm{adj}(J)$:

$$
\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix} -2\lambda\cos\theta & (\pi+2\lambda)\cos\theta \\ -4(1-\sin\theta) & 4(1-\sin\theta) \end{bmatrix}
$$

4. Transformation Matrix $\mathbf{A}$

Based on $\mathbf{A} = \begin{bmatrix} R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y \end{bmatrix}$:

$$
\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1-\sin\theta}}\begin{bmatrix} -2\lambda\cos^2\theta & (\pi+2\lambda)\cos^2\theta \\ -4(1-\sin\theta) & 4(1-\sin\theta) \end{bmatrix}
$$

The determinant is $\det(\mathbf{A}) = \pi R^2$.

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$

$$
\mathbf{G} = \frac{R^2}{4\cos^2\theta(1-\sin\theta)}\begin{bmatrix} 4\lambda^2\cos^4\theta + 16(1-\sin\theta)^2 & -2\lambda(\pi+2\lambda)\cos^4\theta - 16(1-\sin\theta)^2 \\ -2\lambda(\pi+2\lambda)\cos^4\theta - 16(1-\sin\theta)^2 & (\pi+2\lambda)^2\cos^4\theta + 16(1-\sin\theta)^2 \end{bmatrix}
$$

6. Area Element $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

3.13. T_8 to O_8

Let

$$\sin\theta = (2 - x + y)^2 - 1$$
$$\lambda = -\frac{\pi(1 - x)}{2(2 - x + y)}$$

Then

$$x = 1 + \frac{2\lambda}{\pi}\sqrt{1 + \sin\theta}$$
$$y = \left(1 + \frac{2\lambda}{\pi}\right)\sqrt{1 + \sin\theta} - 1$$

Derivation for $x = 1 + \frac{2\lambda}{\pi}\sqrt{1 + \sin\theta}$ and $y = \left(1 + \frac{2\lambda}{\pi}\right)\sqrt{1 + \sin\theta} - 1$

1. Jacobian Matrix of (x, y) with respect to (λ, θ)

The partial derivatives are:

$$x_\lambda = \frac{2}{\pi}\sqrt{1 + \sin\theta}$$
$$x_\theta = \frac{2\lambda}{\pi}\frac{\cos\theta}{2\sqrt{1 + \sin\theta}} = \frac{2\lambda\cos\theta}{2\pi\sqrt{1 + \sin\theta}}$$
$$y_\lambda = \frac{2}{\pi}\sqrt{1 + \sin\theta}$$
$$y_\theta = \left(1 + \frac{2\lambda}{\pi}\right)\frac{\cos\theta}{2\sqrt{1 + \sin\theta}} = \frac{(\pi + 2\lambda)\cos\theta}{2\pi\sqrt{1 + \sin\theta}}$$

The Jacobian matrix is:

$$\frac{\partial(x, y)}{\partial(\lambda, \theta)} = \frac{1}{2\pi\sqrt{1 + \sin\theta}}\begin{bmatrix} 4(1 + \sin\theta) & 2\lambda\cos\theta \\ 4(1 + \sin\theta) & (\pi + 2\lambda)\cos\theta \end{bmatrix}$$

2. Jacobian Determinant

$$\det\left(\frac{\partial(x, y)}{\partial(\lambda, \theta)}\right) = \frac{1}{4\pi^2(1 + \sin\theta)}\left[4(1 + \sin\theta)(\pi + 2\lambda)\cos\theta - 8\lambda(1 + \sin\theta)\cos\theta\right]$$
$$= \frac{4(1 + \sin\theta)\cos\theta(\pi + 2\lambda - 2\lambda)}{4\pi^2(1 + \sin\theta)} = \frac{\pi\cos\theta}{\pi^2} = \frac{\cos\theta}{\pi}$$

3. Inverse Jacobian Matrix $\frac{\partial(\lambda,\theta)}{\partial(x,y)}$

Using $J^{-1} = \frac{\pi}{\cos\theta}\text{adj}(J)$:

$$\frac{\partial(\lambda,\theta)}{\partial(x,y)} = \frac{1}{2\cos\theta\sqrt{1+\sin\theta}}\begin{bmatrix}(\pi+2\lambda)\cos\theta & -2\lambda\cos\theta \\ -4(1+\sin\theta) & 4(1+\sin\theta)\end{bmatrix}$$

4. Transformation Matrix $\mathbf{A}$

Based on $\mathbf{A} = \begin{bmatrix}R\cos\theta\lambda_x & R\cos\theta\lambda_y \\ R\theta_x & R\theta_y\end{bmatrix}$:

$$\mathbf{A} = \frac{R}{2\cos\theta\sqrt{1+\sin\theta}}\begin{bmatrix}(\pi+2\lambda)\cos^2\theta & -2\lambda\cos^2\theta \\ -4(1+\sin\theta) & 4(1+\sin\theta)\end{bmatrix}$$

The determinant is $\det(\mathbf{A}) = \pi R^2$.

5. Metric Tensor $\mathbf{G} = \mathbf{A}^T\mathbf{A}$

$$\mathbf{G} = \frac{R^2}{4\cos^2\theta(1+\sin\theta)}\begin{bmatrix}(\pi+2\lambda)^2\cos^4\theta + 16(1+\sin\theta)^2 & -2\lambda(\pi+2\lambda)\cos^4\theta - 16(1+\sin\theta)^2 \\ -2\lambda(\pi+2\lambda)\cos^4\theta - 16(1+\sin\theta)^2 & 4\lambda^2\cos^4\theta + 16(1+\sin\theta)^2\end{bmatrix}$$

6. Area Element $\sqrt{G}$

$$\sqrt{G} = \sqrt{\det(\mathbf{G})} = \pi R^2$$

4. Advection equation on the sphere

Advection equation:

$$\frac{\partial}{\partial t}q(\lambda,\theta,t) + \nabla\cdot(\boldsymbol{V}q) = 0$$

where $\nabla\cdot(\boldsymbol{V}q)$ is the surface divergence of $\boldsymbol{V}q$

$$\boldsymbol{V} = (u,v), \quad u = u_0(\cos\alpha_0\cos\theta + \sin\alpha_0\cos\lambda\sin\theta), \quad v = -u_0\sin\alpha_0\sin\lambda$$

Based on the transformation, the advection equation on the triangle domain is

$$\frac{\partial}{\partial t}(\sqrt{G}q) + \frac{\partial}{\partial x}(\sqrt{G}u^1q) + \frac{\partial}{\partial y}(\sqrt{G}u^2q) = 0$$

Note that $\sqrt{G}$ is a constant for the current transformation. Hence, the equation can be simplified as

$$\frac{\partial q}{\partial t} + \frac{\partial}{\partial x}(u^1q) + \frac{\partial}{\partial y}(u^2q) = 0$$

Hence, we can solve the problem based on the scheme developed in the previous section.